

CO1-ASSESSMENT TOOL-1

1. Compare two image acquisition systems (CCD vs CMOS) in terms of their impact on image quality in practical applications.
CCD: Charge Coupled Device produces high-quality images with low noise and better light sensitivity.
CMOS: Complementary Metal Oxide Semiconductor uses less power, is faster and is cheaper.
Application: CCD is preferred in medical and scientific imaging, while CMOS is used in smartphones, surveillance cameras.
2. A digital image appears distorted. Explain the causes and suggest improvements.
Possible causes include poor lighting, incorrect camera focus, lens distortion. These affect image formation and reduce image quality.
Improvements include proper illumination, camera calibration, autofocus and distortion correction techniques.
3. Explain how Computer Vision features enable automated surveillance.
Computer vision systems capture images, detect objects, recognize faces, track movement, and analyze activities automatically. These

features enable continuous monitoring, intrusion detection and real-time alerts, making surveillance systems more accurate.

4. Explain how digital image fundamentals and image acquisition improve the accuracy?

High spatial resolution, proper sampling and accurate image sensing help capture detailed land features. Good acquisition methods reduce noise and distortion, improving the accuracy of land classification, vegetation analysis.

5. Explain how image acquisition, sampling and quantization affect OCR performance.

Image acquisition captures the document, sampling determines image resolution and quantization represents pixel intensity. High-quality acquisition, proper sampling and sufficient quantization level produce clear characters, improving OCR accuracy and reducing recognition errors.

6. Explain how sampling and quantization can optimize storage and processing without significantly reducing image quality.

Reduce the sampling rate slightly while maintaining required detail and use an appropriate bit depth such as 8 bits instead of higher values. This decreases file size and processing time while preserving acceptable image quality.

7. Explain how uneven lighting affects image formation and suggest methods to improve image contrast.

Uneven lighting causes non-uniform illumination during image formation, resulting in poor contrast. The problem can be corrected using uniform lighting, histogram equalization, adaptive contrast enhancement or proper camera exposure settings.

8. Explain how improper quantization affects diagnostic accuracy in medical imaging.

Low quantization levels reduce intensity resolution, causing fine details and small intensity differences to disappear. This may hide important abnormalities, reducing diagnostic accuracy. Higher bit-depth quantization preserves more image formation.

9. Explain how image acquisition and computer vision features support reliable decision making.

High-quality cameras capture clear road images, while Computer Vision detects lanes, traffic signs, pedestrians and vehicles. Accurate image acquisition and feature extraction help the vehicle make safe and reliable driving decisions.

10. Explain how sampling rate and intensity resolution help detect fine details in images. A high sampling rate provides better spatial resolution, while high intensity resolution captures small brightness variations. Together, they preserve fine details, making object detection and analysis more accurate.